11–TRIM

11.1 DEFINITION OF TRIM

Trim (T) is the difference between fore and aft draughts. It is a state of equilibrium in the longitudinal direction. There are three possible states of trim that a ship may assume.

11.1.1 – Even Keel:

When the draughts forward and aft are the same a vessel is said to be on **Even Keel**. i.e.

$$d_F = d_A$$

and: $$T = 0$$

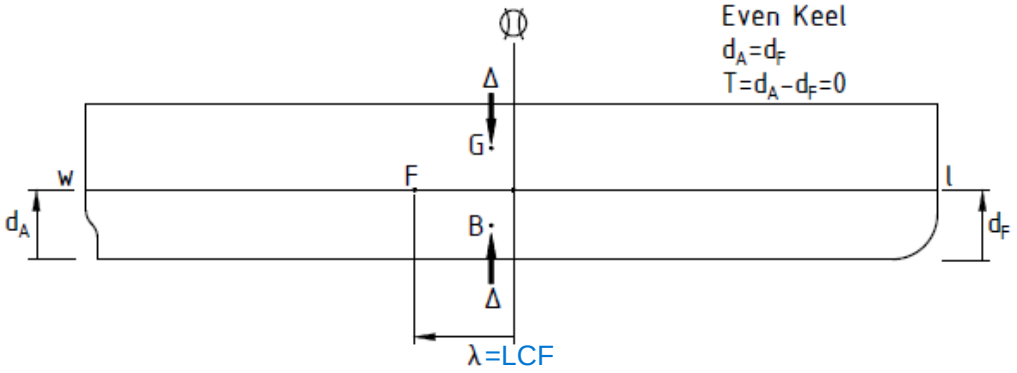


Figure 11.1 – Even Keel

11.1.2 – Head Trim:
A ship is said to have a **Head Trim** when the draught forward is greater than the draught aft i.e.

$$d_F > d_A$$

and: $$T = d_F - d_A$$

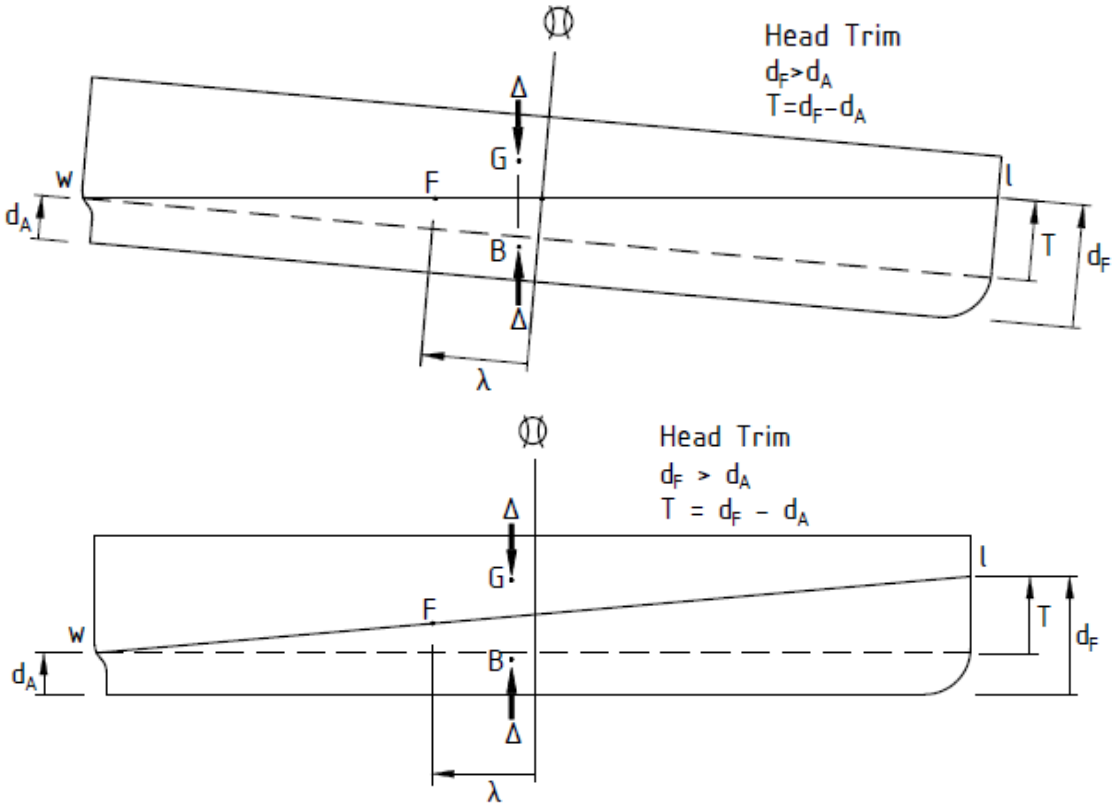


Figure 11.2 – Head Trim

11.1.2 – Stern Trim:

When the draught aft is greater than the draught forward a ship is said to have a **Stern Trim** i.e.

Suitable trim for stability $d_A > d_F$

and: $T = d_A - d_F$

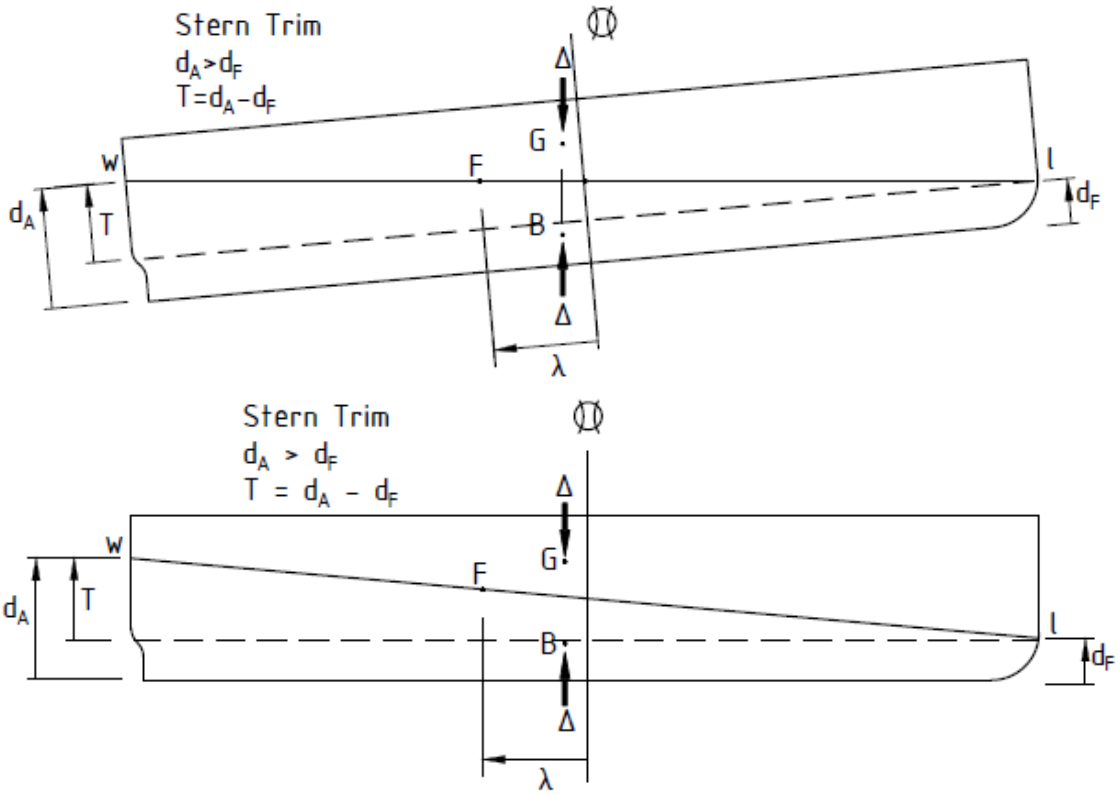


Figure 11.3 – Stern Trim

11.2 LONGITUDINAL METACENTRE (M_L)

It is the point of intersection of the vertical line through the centre of buoyancy of a vessel when upright (even keel in this case) and the vertical line through the centre of buoyancy when inclined. It may be considered as the instantaneous centre of the locus of the longitudinal centre of buoyancy.

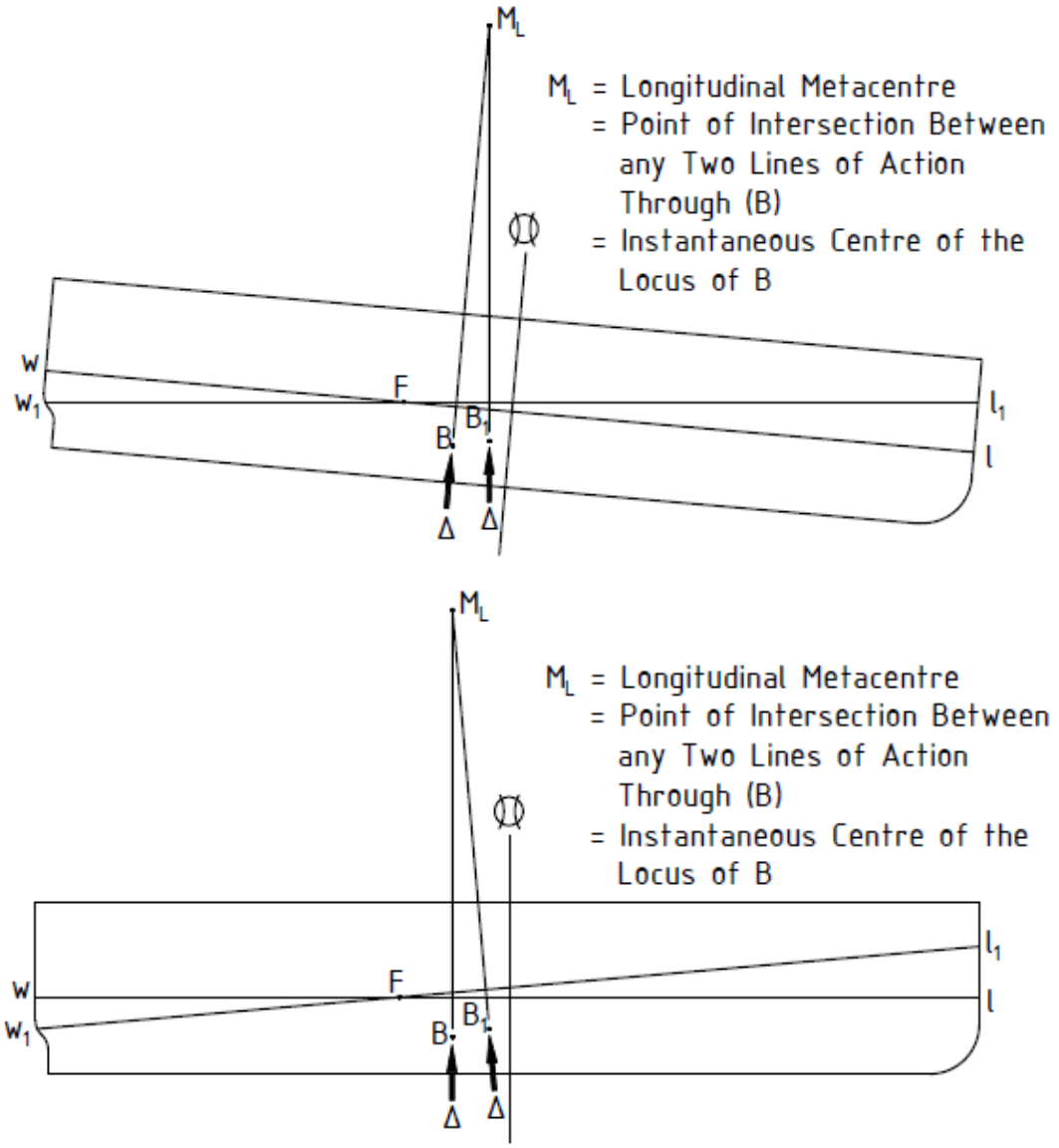


Figure 11.4 – Longitudinal Metacentre

11.2 TRUE MEAN DRAUGHT

The draught of a ship taken at midships is the average of the fore and aft draughts. This referred to as the **Arithmetic Mean Draught** (d_M).

$$d_M = \frac{d_F + d_A}{2}$$

Midship is however not the neutral axis about which a vessel changes trim. Trim changes about the centre of flotation (F). The axis through F is therefore the neutral axis and hence the draught at F will remain constant for any particular displacement. This draught is referred to as the **True Mean Draught** (d_T).

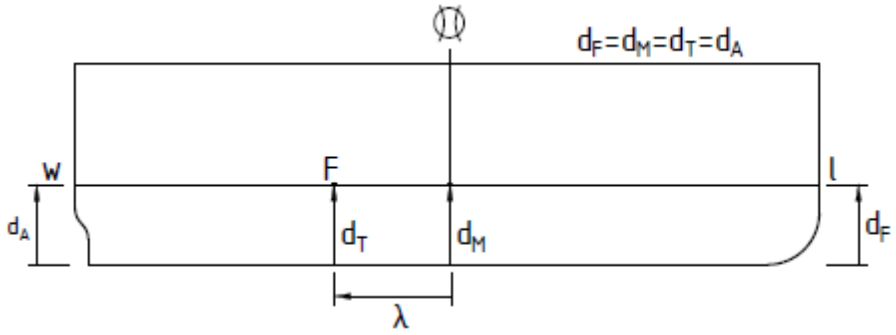


Figure 11.5

On even keel: $d_T = d_M = d_F = d_A$

When however, a ship is trimmed by the head as in Figure 11.6 and (F) is aft of midship:

$$d_F > d_M > d_T > d_A$$

and:

$$d_T = d_M - x$$

When however, (F) is forward of midship:

$$d_F > d_T > d_M > d_A$$

and:

$$d_T = d_M + x$$

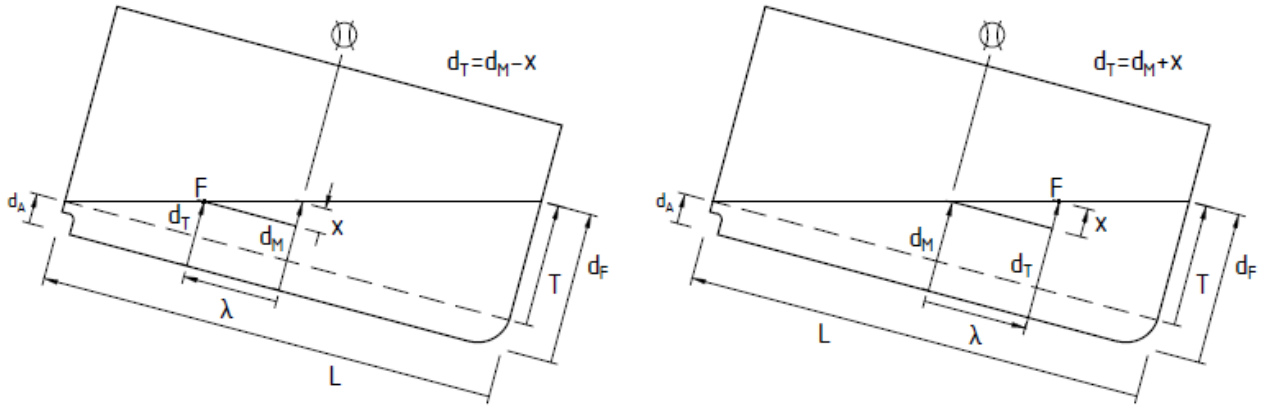


Figure 11.6

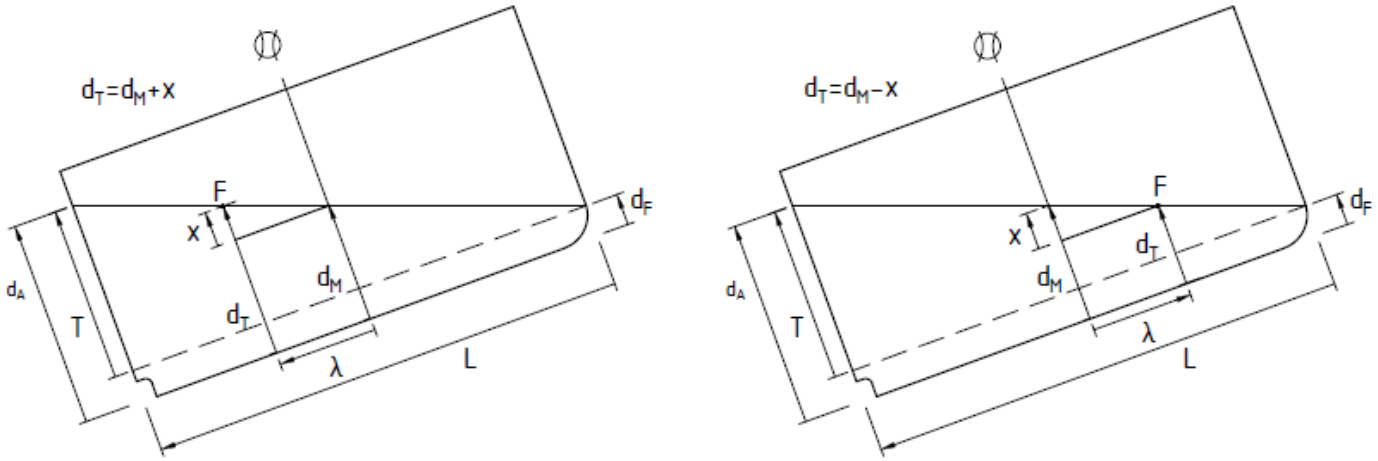


Figure 11.7

When now a ship is trimmed by the stern as in Figure 11.7 but with (F) aft of midship:

$$d_F < d_M < d_T < d_A$$

and:

$$d_T = d_M + x$$

And when (F) is forward of midship:

$$d_F < d_T < d_M < d_A$$

and:

$$d_T = d_M - x$$

Therefore, in summary:

$$d_T = d_M \pm x$$

Now by similar triangles:

$$\frac{x}{\lambda} = \frac{T}{L}$$

Therefore:

$$x = \frac{T\lambda}{L}$$

And:

$$d_T = d_M - \frac{T\lambda}{L}$$

And since d_M is halfway between perpendiculars, then:

$$d_M = \left(\frac{d_F + d_A}{2}\right)$$

11.3 EFFECT OF SHIFTING SMALL MASS ON END DRAUGHTS

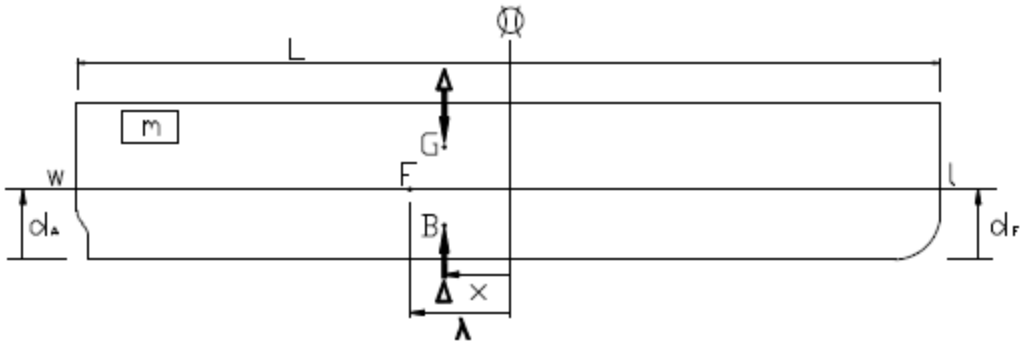


Figure 11.8

Consider a vessel initially on even keel at the waterline (wl) to have a small mass (m) positioned in it as shown in Figure 11.9. The mass being part of the displacement. Under this condition B and G will be at the same longitudinal distance (x) from midship and the vessel is in equilibrium. Trim:

$$T = d_F - d_A = 0$$

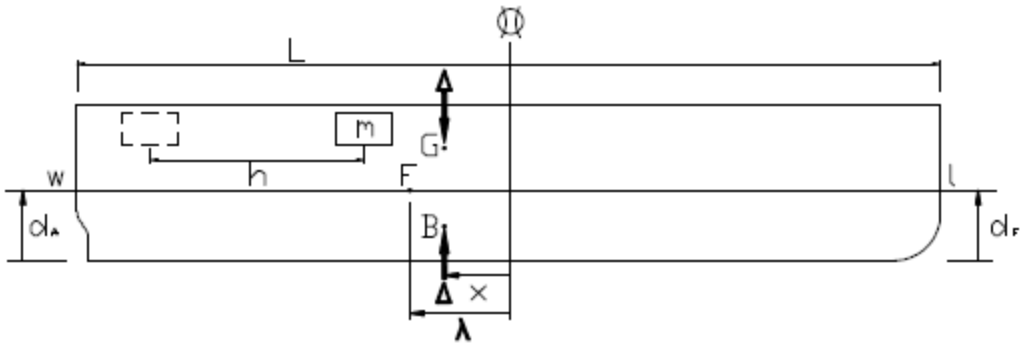


Figure 11.9

Now move (shift) the mass longitudinally forwards a distance (h) as in Figure 11.10.

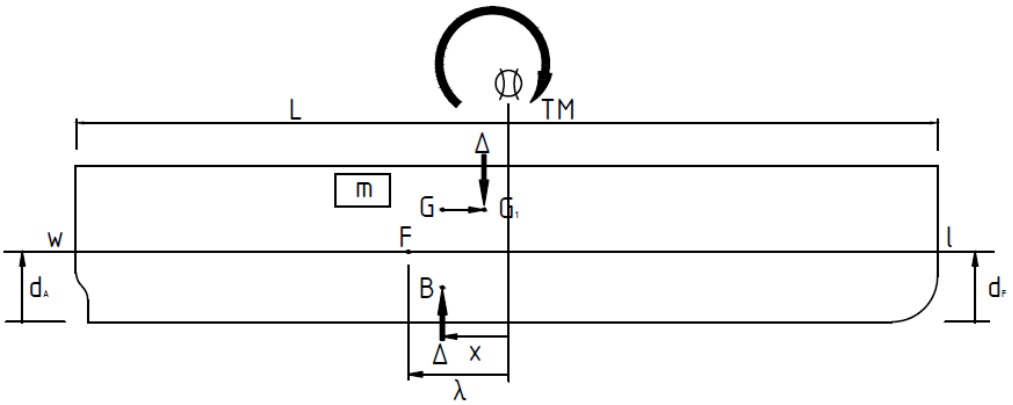


Figure 11.10

This movement will cause:

1. G to shift to G_1 as in Figure 11.11, temporarily leaving B behind in its position.

$$GG_1 = \frac{mh}{\Delta}$$

2. A trimming moment (TM) will then come into action about the centre of flotation (F).

$$TM = \Delta . GG_1$$

3. With trimming taking place, the underwater volume changes form thereby causing the centre of buoyancy to also shift in the trimming direction. A counter TM is hence generated by the force of buoyancy until at the position B_1, it will cancel out the TM caused by displacement force to again bring the vessel back into equilibrium but at a new waterline (w_1l_1) as shown in Figure 11.11.

   At this stage the final draughts fore and aft will be:

$$d_{F1} = d_F + Change\ in\ draught\ forward$$

   And

$$d_{A1} = d_A - Change\ in\ draught\ aft$$

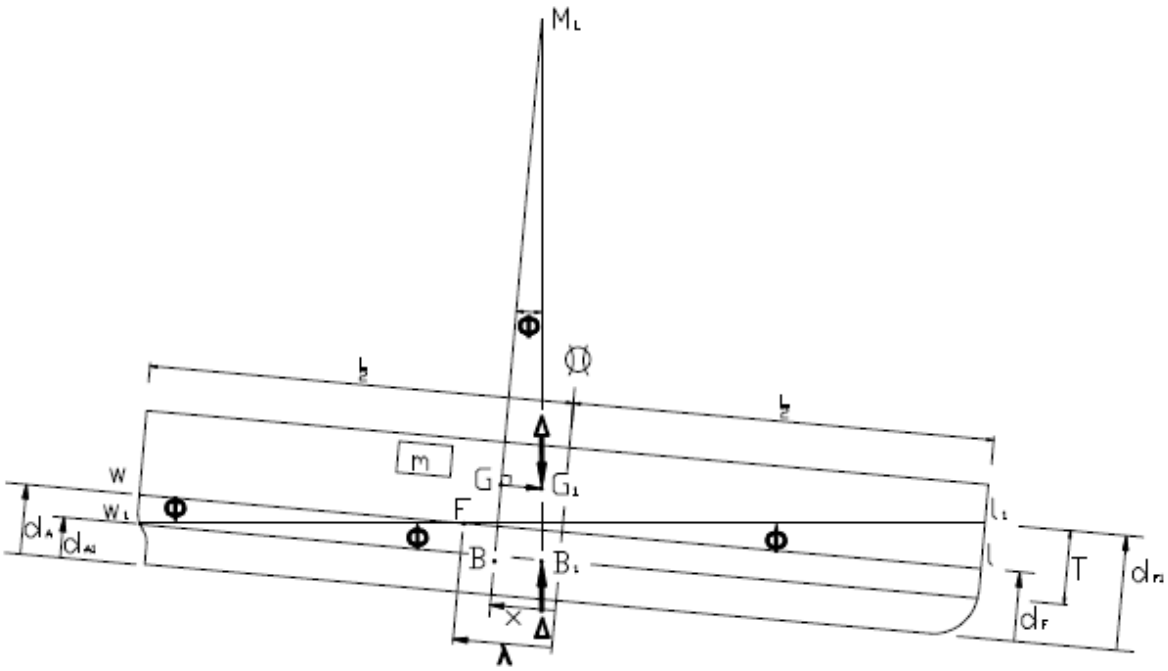


Figure 11.11

Trim will now be T_1

And change in trim:

$$t = T_1 - T = T_1 - 0 = T_1$$

Assuming the ship to be wall sided at the ends:

Then from similar triangles: GG_1M_L, ll_1F and ww_1F:

$$\frac{T_1}{L} = \frac{GG_1}{GM_L} = \frac{ll_1}{lF} = \frac{ww_1}{wF}$$

From which:

$$GG_1 = \frac{T_1.GM_L}{L}$$

And hence:

$$TM = \Delta.GG_1 = \Delta\frac{T_1.GM_L}{L} = \Delta.\frac{t.GM_L}{L}$$

If therefore a TM = $[\Delta.GM_L \times \frac{t}{L}]$ is required to change trim by (t) m, then the '**M**oment required to **C**hange **T**rim by **1** m':

$$MCT1_m = \Delta\frac{1.GM_L}{L} = \frac{\Delta.GM_L}{L}$$

From which it can be deduced that:

$$t = \frac{TM}{MCT1_m}$$

Now:

$$Change\ in\ draught\ forward = ll_1 = \frac{T_1}{L} \times lF = \frac{t}{L} \times lF$$

And:

$$Change\ in\ draught\ aft = ww_1 = \frac{T_1}{L} \times wF = \frac{t}{L} \times wF$$

Therefore:

$$d_{F1} = d_F + \frac{t}{L} \times wF = d_F + \left(\frac{t}{L}\left(\frac{L}{2} + \lambda\right)\right)$$

And, final draught aft:

$$d_{A1} = d_A - \frac{t}{L} \times lF = d_F - \left(\frac{t}{L}\left(\frac{L}{2} - \lambda\right)\right)$$

Now since trim change may be by the head or stern, and since the position of F may be forward or aft of midship, then by extension the final draughts due to trim change may be expressed as:

$$d_1 = d \pm \left(\frac{t}{L}\left(\frac{L}{2} \pm \lambda\right)\right)$$

11.4 EFFECT OF ADDING OR REMOVING A SMALL MASS ON END DRAUGHTS

When a small mass (m) is ***ADDED*** to a vessel initially of displacement Δ the following actions will occur progressively.

1. Displacement will increase to Δ_1, and
2. A trimming moment will come into action to tilt the vessel to final draughts d_{F1} and d_{A1} as shown finally in Figure 11.15.

For ease of determining these new draughts however, it may be conveniently assumed that the mass is first added directly at F and thereafter shifted to its final resting place. The actions may therefore be separated into:

1. Addition of the mass as in Figure 11.12.

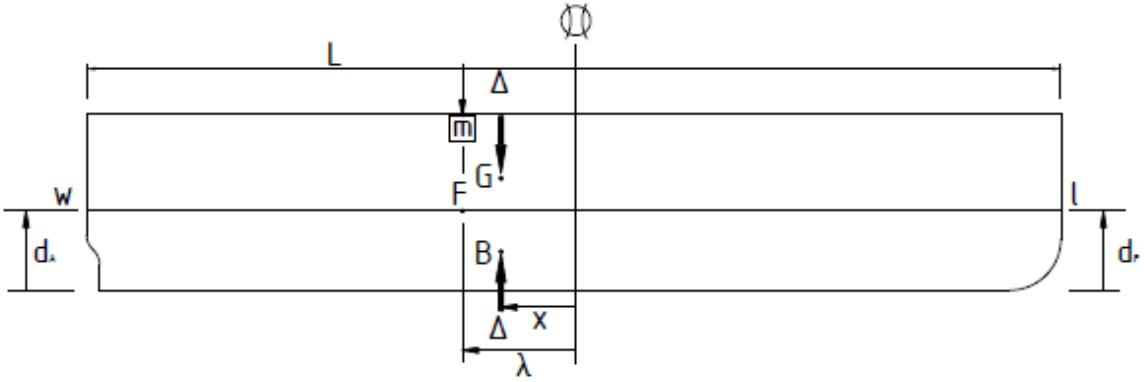


Figure 11.12

2. Parallel sinkage (s) which submerges the vessel to an intermediate waterline w_1l_1 and displacement Δ_1 with centre of flotation at F_1 as in Figure 11.13. The intermediate draughts at this stage will be (d_F+s) fwd and (d_A+s) aft. B will shift upwards and to the left to B_0 leaving G behind. The sinkage may be calculated from:

$$s = \frac{m}{TPC}$$

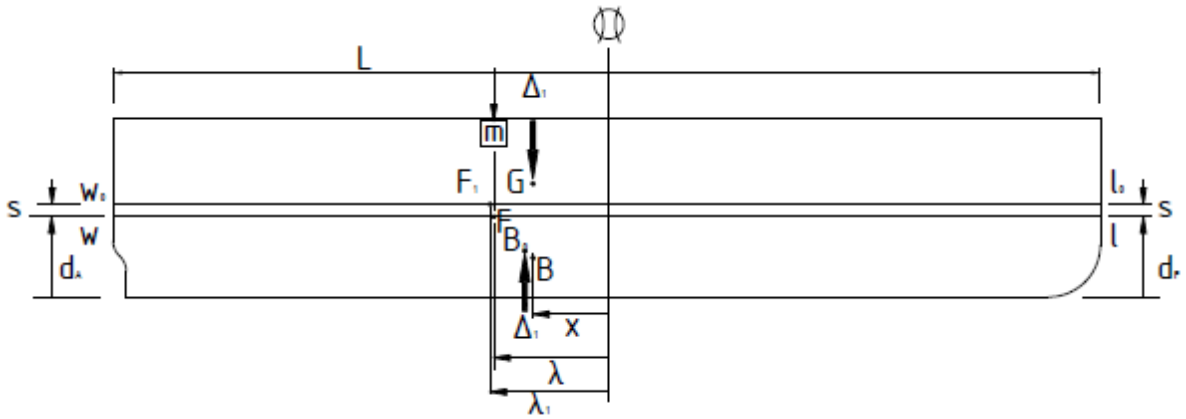


Figure 11.13

3. Shifting of the mass a distance (a) to its final resting place. This will cause G to shift to G_1 thereby causing a trimming moment (TM) to come into action as in Figure 11.14.

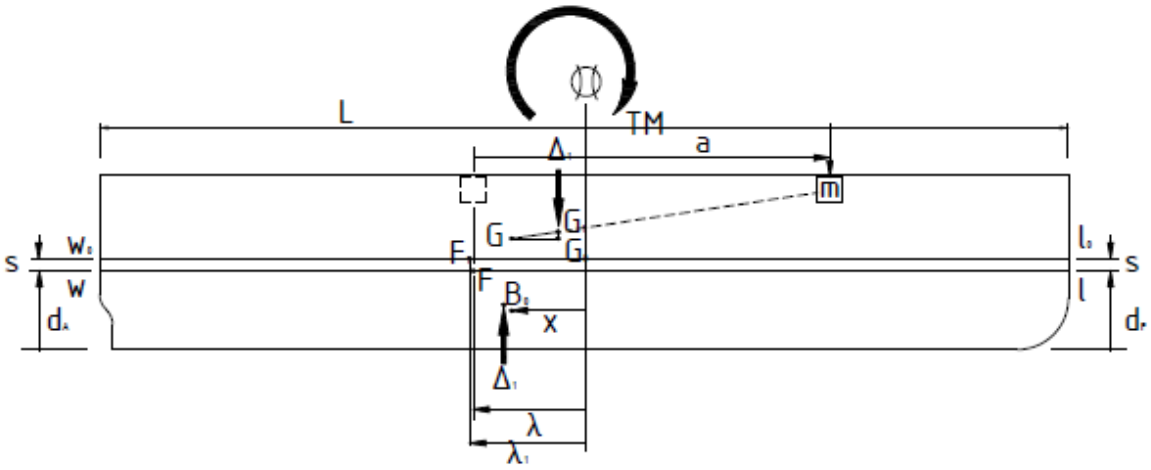


Figure 11.14

4. Tilting action that brings the vessel to its final draughts as shown in Figure 11.15. The trimming moment:

$$TM = \Delta_1 . GG_1$$

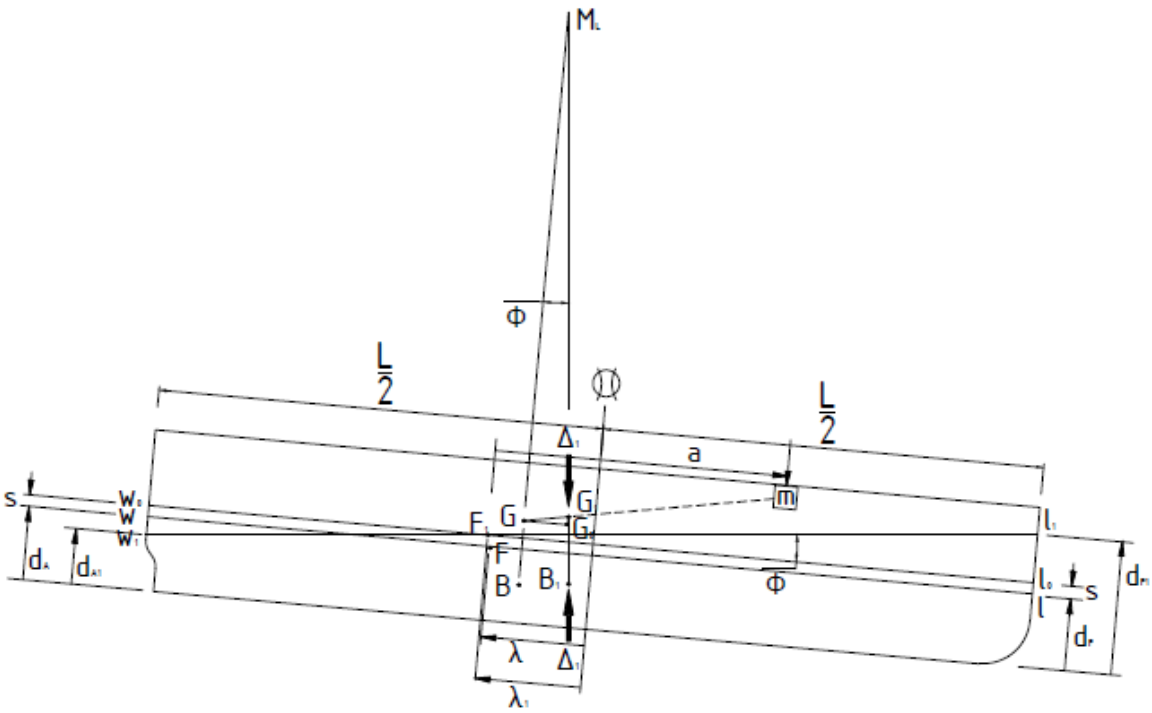


Figure 11.15

On assumption that the effect of addition of the small mass does not change the ship's hydrostatic data significantly, then $(\lambda_1 \cong \lambda)$.

The final draughts will then be:

$$d_{F1} = d_F + s + ll_1 = d_F + s + \left(\frac{t}{L}\left(\frac{L}{2} + \lambda\right)\right)$$

$$d_{A1} = d_A - s - ww_1 = d_A - s - \left(\frac{t}{L}\left(\frac{L}{2} - \lambda\right)\right)$$

When a small mass (m) is ***REMOVED*** from a vessel the actions described will be in reverse and sinkage will be considered as negative.

Therefore, in general:

$$d_1 = d \pm s \pm \left(\frac{t}{L}\left(\frac{L}{2} \pm \lambda\right)\right)$$

11.6 EFFECT OF ADDING OR REMOVING LARGE MASS ON END DRAUGHTS

Figure 11.16 shows a ship initially in equilibrium at a waterline (wl) with centres of buoyancy and flotation at B and F respectively.

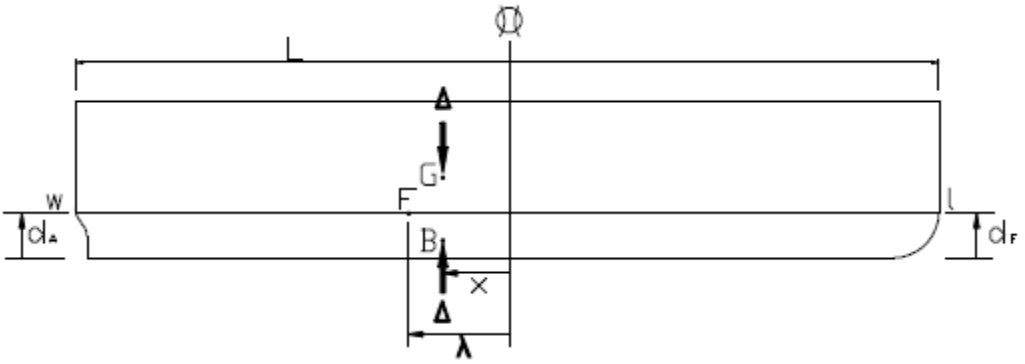


Figure 11.16

Now ***ADD*** a large mass (M) as shown in Figure 11.17.

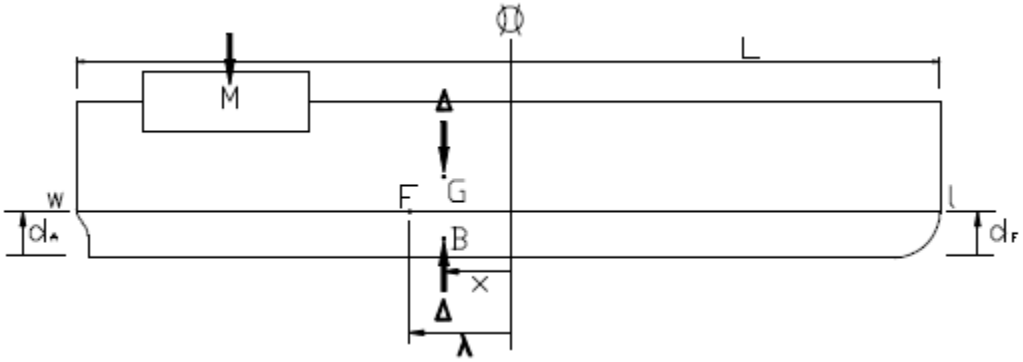


Figure 11.17

Due to the above action displacement will increase to Δ_1 and the vessel will float at a new waterline w_1l_1 with assumed true mean draught d_T. Its hydrostatic data will alter significantly to cause F to shift to F_1 and B to B_1 as in Figure 11.18.The centre of gravity will also shift from G to G_1.

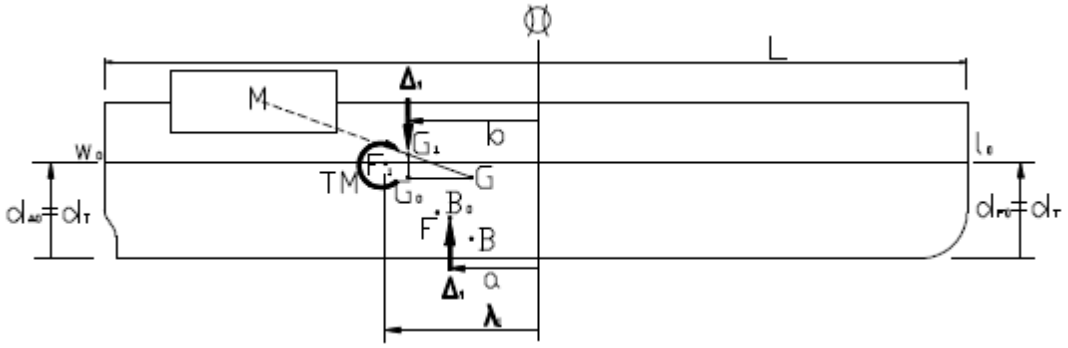


Figure 11.18

If under this condition the longitudinal distance FG_1 and FB_1 are the same then the final draught will settle at d_T as assumed.

If on the other hand they end up not being the same then a TM will come into action to tilt the vessel until it again attains equilibrium as in Figure 11.19, by which stage B_1 will have finally shifted at B_2 to make FG_1 equal to FB_2.

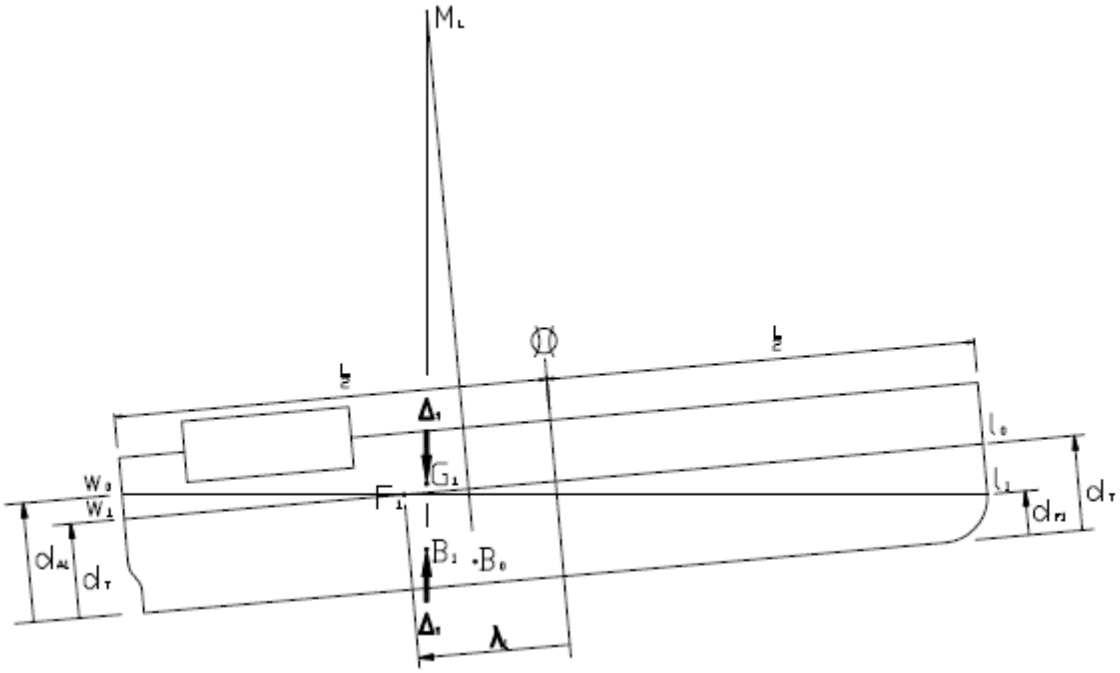


Figure 11.19

The final draughts will then be:

$$d_{F1} = d_T - ll_1 = d_T - \left(\frac{t}{L}\left(\frac{L}{2} + \lambda\right)\right)$$

$$d_{A1} = d_T + ww_1 = d_T + \left(\frac{t}{L}\left(\frac{L}{2} - \lambda\right)\right)$$

When a large mass (M) is ***REMOVED*** from a vessel the actions described will be in reverse.

Therefore, in general:

$$d_1 = d_T \pm \left(\frac{t}{L}\left(\frac{L}{2} \pm \lambda\right)\right)$$

11.7 CHANGE IN END DRAUGHTS DUE TO CHANGE IN DENSITY

When a ship sails from sea water into fresh or river water or vice versa without change in displacement, there will be change in displaced volume resulting in positive or negative sinkage (s) as the case may be.

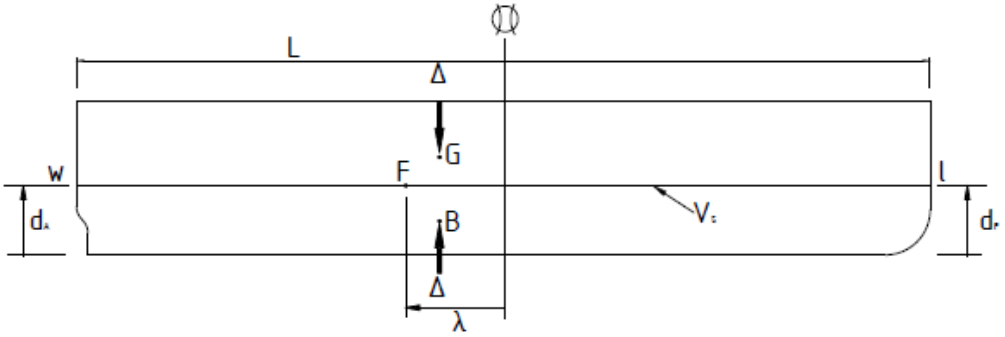


Figure 11.20

Consider the vessel in Figure 11.20 to have a displaced volume ∇_S with centre of flotation at F. When she moves from sea water of density ρ_S into fresh water of density ρ_F the volume will increase to ∇_F causing it to sink to a new waterline w_1l_1. The centre of flotation will then shift to F_1 as in Figure 11.21.

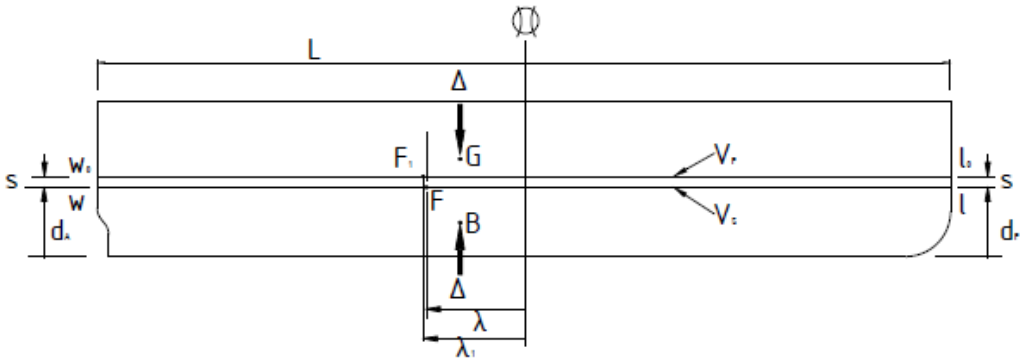


Figure 11.21

If the change in waterplane area A_W is considered negligible then,

$$A_{WS} = A_{WF} = A_W$$

and hence LCF_S and LCF_F may also be considered to be the same. Assuming sinkage to be parallel, B will then shift B_0 as in Figure 11.22, From this, increase in volume will be equal to the waterplane area multiplied by the sinkage:

$$A_W \times s = \nabla_F - \nabla_S = \frac{\Delta}{\rho_F} - \frac{\Delta}{\rho_S}$$

And:

$$s = \frac{\Delta}{A_W}\left(\frac{\rho_S - \rho_F}{\rho_F \rho_S}\right) \quad \text{(m)}$$

Or:

$$s = \frac{\Delta}{A_W}\left(\frac{\rho_S - \rho_F}{\rho_F \rho_S}\right) \times 100 \quad \text{(cm)}$$

Or:

$$s = \frac{\Delta}{TPC_S}\left(\frac{\rho_S - \rho_F}{\rho_F}\right)$$

Now for ships the centres of buoyancy usually move aftwards as draught increases. Therefore, F will move longitudinally to F_1. This will cause the displacement moment to become greater than the buoyancy moment thereby bringing into action a forward TM. The vessel will then trim until it again attains equilibrium when B_0 have finally shifted to B_1 as in Figure 11.23. At this stage longitudinal distances F_1G and F_1B_2 will be equal to each other.

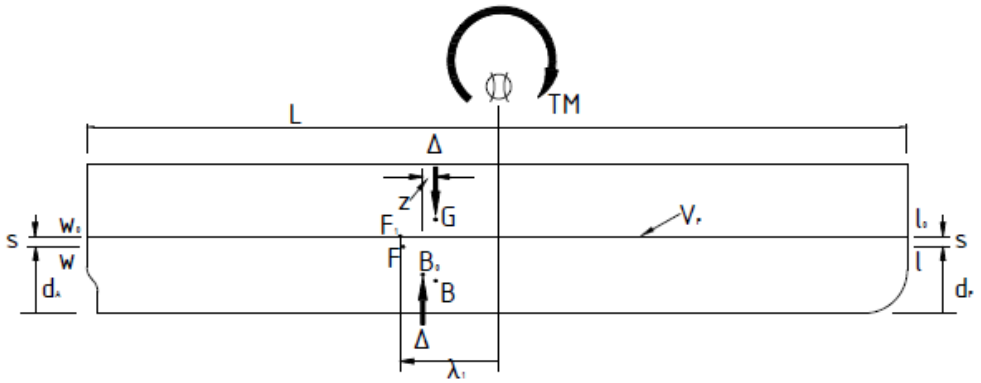


Figure 11.22

By taking moments of volume longitudinally about B:

$$(\nabla_F \times z) = (\nabla_S \times 0) + (\nabla_F - \nabla_S)(F_1 B)$$

Assuming the positions F1 and F are not significantly different, then:

$$z = \frac{\left(\Delta/\rho_F - \Delta/\rho_S\right)(FB)}{\Delta/\rho_F} = \left(\frac{\rho_S - \rho_F}{\rho_S}\right)(FB)$$

And:

$$TM = \Delta \times z = \Delta\left(\frac{\rho_S - \rho_F}{\rho_S}\right)(FB)$$

The final draughts will then be:

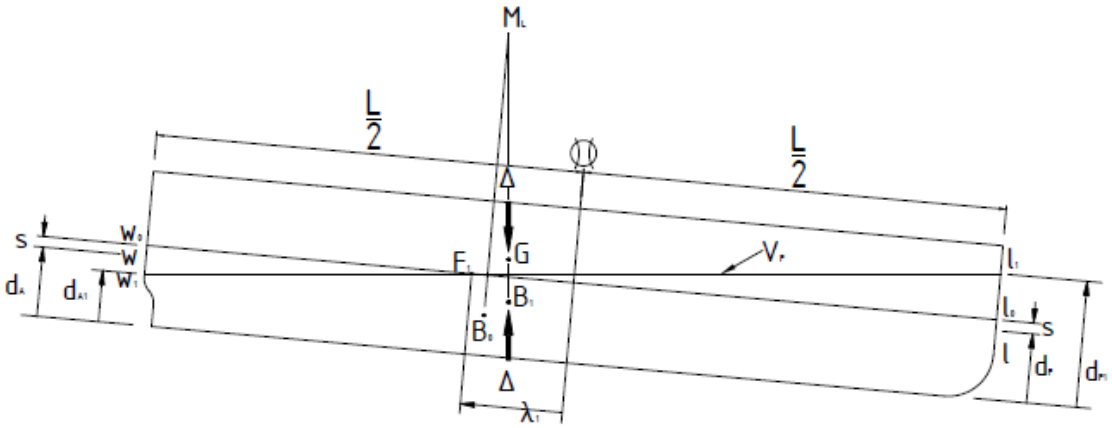


Figure 11.23

$$d_{F1} = d_F + s + ll_1 = d_F + s + \left(\frac{t}{L}\left(\frac{L}{2} + \lambda\right)\right)$$

$$d_{A1} = d_A + s - ll_1 = d_F + s - \left(\frac{t}{L}\left(\frac{L}{2} - \lambda\right)\right)$$

Or in general:

$$d_1 = d \pm s \pm \left(\frac{t}{L}\left(\frac{L}{2} \pm \lambda\right)\right)$$

11.8 DETERMINATION OF LCG FROM OBSERVED DRAUGHTS

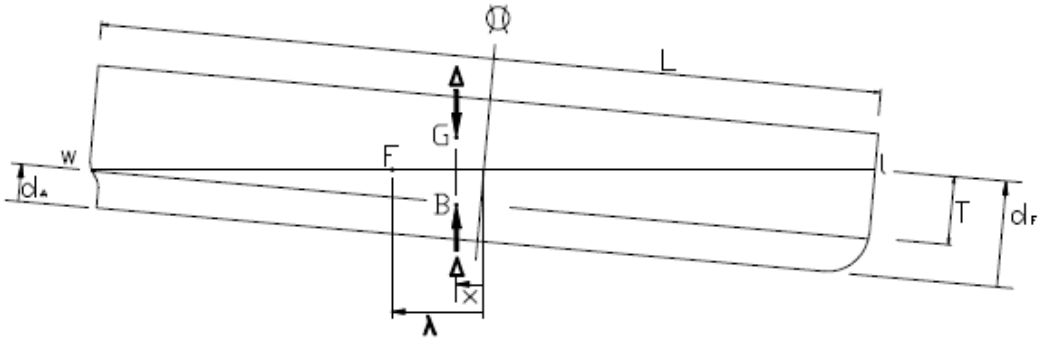


Figure 11.24

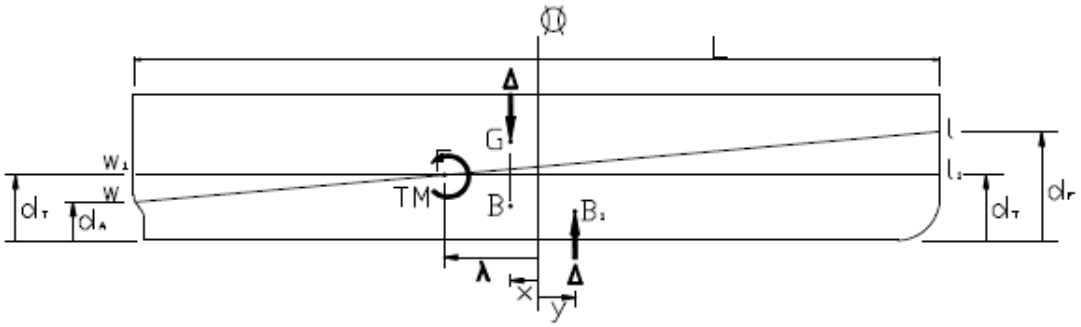


Figure 11.25

Consider the vessel in Figure 11.24 to be floating in equilibrium at waterline wl with its centre of gravity G which is yet to be determined to be at a distance (x) from midship. Her trim is T and its centre of buoyancy will be at B.

If now the vessel is imagined to be brought onto even keel waterline w_1l_1 by a trimming moment TM, the position of the centre of buoyancy will shift to B_1, a distance (y) from midship.

Now from:
$$t = \frac{TM}{MCT}$$

$$TM = T \times MCT$$

$$\Delta(y + \lambda) - \Delta(\lambda - x) = T \times MCT$$

From which:
$$x = \frac{T \times MCT}{\Delta} - y$$

Or in general, depending on the positions of B1 and G relative to midship and F:

$$x = \pm y \pm \frac{T \times MCT}{\Delta}$$